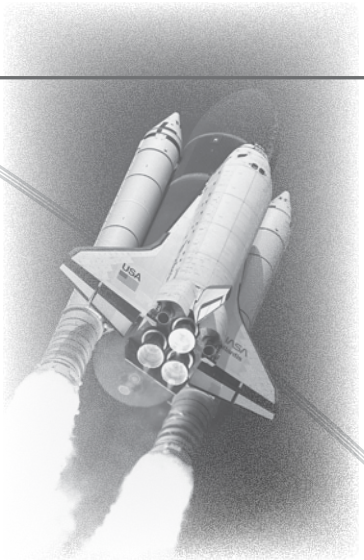
CHAPTER 8



THE NEW LANDSCAPE

An operational TDRSS did not mean that a GN was not needed. It still was, just not in the same way as before. Phase out from the STDN organization did not put an end to NASA's ground station activities. Many sites operated like they did before TDRSS came along, only now they did so for different reasons. No longer called the STDN, the GN played a different role to support a different mission.

In the Pacific, the Kauai Station supported the University of Hawaii specifically, and the Earth science community at large, operating as the Kokee Park Geophysical Observatory (KPGO). As far back as 1981, operations at Kauai had reduced from 24/7 around-the-clock operations to a standard eight-hour, five-days-a-week schedule. Like many of the stations in the network, Hawaii had seen its fair share of "close calls" when it came to closing. Originally scheduled for complete phase out in April of 1984, it kept getting postponed while NASA awaited TDRSS to come online.[1]

The original plan was to transfer the equipment and tracking responsibilities of the station to the Navy's Pacific Missile Range Facility (PMRF) on nearby Barking Sands. The memorandum of agreement for this transfer had in fact been signed-off by both NASA and the DOD when the *Challenger* accident happened in January 1986. After STS-26, the station was

officially closed on 30 September 1989 after TDRS-4 was checked out and declared operational, and the transfer to PMRF took take place at that time. The Navy took possession of most of the equipment with the exception of one key asset which NASA retained, the 9-meter (30-foot) S-band antenna system used by KPGO.

One of KPGO's first assignment in its new Earth observation role was to support Goddard's Crustal Dynamics Project. It joined several other observatories in the continental United States, Japan, Chile, and Australia to make ultra-precise position determination of the crust using the tracking measurement technique called Very Long Baseline Interferometry, or VLBI. Kokee Park participated in these NASA and Naval Observatory sponsored experiments and because of its location in the mid-Pacific tectonic plate, was among the most active of the more than 30 observatories around the world.[2]

In this application of VLBI, several radio telescopes (observatories) simultaneously received signals from extra-galactic quasar radio wave sources. Using lasers and the most precise clocks in the world, the difference in time of arrival of the signals due to the slightly different path lengths from the quasar to each VLBI observatory around the world could be determined to an accuracy of $1x10^{-11}$ seconds (10-trillionth of a second) and their relative positions measured to better than 1 centimeter (0.4 inches).[3] These ultra-precise position measurements, when made repeatedly over several years to decades, allowed scientists to plot the contemporary motion of the tectonic plates—the enormous pieces of Earth's crust—as they moved slowly with respect to each other. The observatories were also able to monitor other geodynamic parameters such as the very complex variation of Earth's spin rate with the minute wobble of the spin axis.

All these gave insight, unavailable before this time, into global geophysics and the underlying forces that led to earthquakes, for example. VLBI measurements which have been made at KPGO since 1984 and which continue today, showed clearly that the Hawaiian Islands (located on the Pacific plate) move at a rate of some 9 centimeters (3.5 inches) a year with respect to the North American plate.

In this new mission, Kokee Park was a principal ground station that diversified to support science application satellites from across several U.S. government agencies. One was the Department of Commerce's PEACESAT (Pan-Pacific Education and Communication Experiments by Satellite) program, which provided medical, educational, and cultural satellite communications between Hawaii and the remote islands in the Pacific basin. It also supported GSFC's Interplanetary Monitoring Platform-8 (IMP-8) that monitored Earth's magnetic field and solar wind activities. In addition, KPGO supported the GOES (Geostationary Observational Environmental Satellite) program for the state of Hawaii.

In South America, the station at Santiago—one of the original Minitrack sites established in 1957 and which had been mostly operated by

the Chileans—was completely turned over in 1988 to the University of Chile, who operates it to this day. NASA, though, still has a stake in the station, but now strictly as a customer. Because of its optimal location in the Southern Hemisphere, the United States pays Chile about half a million dollars a year to support a finite number of satellite passes. The number averages out to about two or three passes a day depending on what missions need support. Bill Watson at Headquarters explained how the Agency uses the station today to meet its data pass requirements.

> Some days we take none, some days we take a lot depending on what's going on. Sometimes Santiago is one of a few Southern Hemisphere stations that we have so when something is happening in the Southern Hemisphere and we need coverage, that is a convenient place. Sometimes there are planetary flybys, JPL satellites whizzing by the Earth. They are going so fast near the Earth that their big antennas can't slew fast enough to track, so stations like Santiago will support it.[4]

In effect, a station that the United States in cooperation with the government of Chile started nearly 50 years ago continues in its legacy today.

The surest sign that the era of NASA's world-wide network of spaceflight tracking stations have come and gone was when Bermuda was finally phased out in 1997. Since 1962, when it first gave John Glenn the "go for orbit" call, Bermuda had supported every human spaceflight that NASA had flown, making the critical go/no-go call on all of them—an impressive resume of 118 missions. On 19 November 1997, *Columbia* took to the air on the 88th flight of the Shuttle program. In a Space Shuttle first, the entire stack was rolled from its usual belly-up to a belly-down position in a 40-second Roll-To-Heads-Up (RTHU) maneuver six-minutes after liftoff. Prior to this flight, such a maneuver would have been used only if a Trans-Atlantic Abort emergency landing were declared by Mission Control due to a failed main engine or the loss of cabin pressure during the crew's ascent into orbit.[5]

The RTHU maneuver was added to eliminate the Shuttle's large External Tank from obstructing the communication line-of-sight between the vehicle's antennas and the TDRS-East spacecraft. By doing so, a smooth handover from Merritt Island to TDRSS could be made with only a momentary gap in coverage. Up until that mission, the Shuttle switched over to the space-based tracking satellites only after reaching orbit some eight and a half minutes after launch. The RTHU maneuver—used ever since on all low inclination, easterly launches—allowed the Orbiter to communicate with TDRS about two and a half minutes sooner. (Higher inclination launches towards the northeast for flights to the ISS did not have to perform the roll maneuver due to the availability of DOD tracking stations along the East Coast.) Although tricky, the roll

maneuver did not unduly stress the vehicle since it was done well after the SRBs had jettisoned and the Shuttle itself had passed through the thickest part of the atmosphere so that aerodynamic stresses were not a problem.[6]

Bermuda was needed no more. Ultimately though, the decision to close the site came down to cost. The closing saved NASA $5 million a year, which coincidently, was the same amount it cost to build the station in 1961.[7]

With Bermuda closed, Merritt Island/Ponce de Leon (MILA/PDL) became the only source of tracking data for the first seven minutes of each Space Shuttle launch. Despite the phase out of all the original ground stations in the STDN, MILA still remains. In fact, it is as essential today after over 100 Space Shuttle launches as it was for STS-1 back in 1981.

Located adjacent to Launch Complex 39 at the KSC, MILA (acronym for "Merritt Island Launch Annex to Cape Canaveral," the early name of the area that was eventually renamed the John F. Kennedy Space Center) was greatly expanded in 1972 right after Apollo 17. The site was used to get Shuttle data to the Launch Control Center at Kennedy during prelaunch testing and terminal countdown. Once the vehicle cleared the tower, MILA transmitted data to Mission Control in Houston. The GSFC first established MILA in 1966 as a primary MSFN station to provide Earth orbit support for Apollo. The station received the first television signals using Unified S-Band during the Apollo Saturn 203 mission on 5 July 1966 on a flight first testing the performance of the liquid hydrogen fuel in the S-IVB third stage to verify its on-orbit restart capability.

Shortly thereafter, GSFC worked with JSC and equipped the station with a complete set of flight control consoles in order to train Mission Control engineers during prelaunch testing of the CSM and LM. The consoles were used until the end of the program in December 1972. In the mid-1970s when S-band transmitters were added to NASA's Delta and Atlas-Centaur expendable launch vehicles, MILA became really busy, supporting those programs as well as Skylab and Apollo-Soyuz. When the STADAN station at Fort Myers was shut down in 1972, its VHF telemetry and communication equipment were relocated to MILA, greatly enhancing the station's capability to also support application satellites programs.[8] With 13 antennas, including a 9-meter (30-foot) USB system, C-band radar, full TT&C capabilities and a UHF air-to-ground voice link for backup, MILA was (and is) NASA's primary launch area tracking station.[9]

As development of the large SRBs of the Space Shuttle neared completion in the mid to late 1970s, GSFC, working with the MSFC, predicted a potential "plume attenuation" problem in which the high temperature, highly reflective plasma in the rockets' exhaust interfered with MILA's reception of signals from the Shuttle early in its ascent. The phenomenon would have been something akin to trying to follow the flight of a bird with a pair of binoculars while looking through a cloud. To solve this problem, a site with a different



Prior to STS-87, Shuttle flights on easterly trajectories went all the way into orbit on their backs. The Shuttle now performs a Roll-to-Heads-Up (RTHU) maneuver prior to main engine cutoff so that communication with TDRSS can be established some two and a half minutes sooner. This allowed the Bermuda Station to be closed down in 1997. (NASA Image Number GPN-2000-000736)

look-angle had to be found. What followed was the Ponce de Leon Station (PDL) "wing-site" that was set up in 1979. Located 64 kilometers (40 miles) north of MILA on 1.4 acres of U.S. Coast Guard property, it was just south of the Ponce de Leon Inlet at New Smyrna Beach. PDL provided a different viewing angle, putting it outside of the "plume shadow." A 4.3-meter (14-foot) USB system was setup specifically to circumvent this problem.

Upon loss-of-signal at MILA, PDL took over as the primary station during a launch, communicating with the Shuttle during its second minute of flight. PDL, however, could not directly communicate with Mission Control at the JSC; MILA still had to do this. Therefore, a three-hop, microwave system with towers at Shiloh and North Wilson were built to relay data from the wing-site to the main location (again, not unlike relaying of cell phone calls). Strictly a supplement to Merritt Island, Ponce de Leon was normally not even staffed, with two or three technicians dispatched to the station to support flight readiness, countdown activities and the actual launch.[10]

With PDL tagged to cover this 60-second gap, according to Shuttle flight rules, a backup to the site itself had to be identified. This dual-redundant

requirement harbored back to the early days of NASA human flight operations where a back up was required for any system designated as primary. To this end, a search was conducted in the southern Florida area to find a location suitable to back up Ponce de Leon. Communication link analysis showed that the Air Force's Jonathan Dickinson Missile Tracking Annex (JDMTA) some 150 kilometers (95 miles) south of the Cape near Jupiter, Florida, could back up PDL for S-band downlink. The DOD had constructed this facility in 1985 and 1986 on 11 acres of land in the state park to provide launch support for their launches and missile testing activities. This allowed the Air Force to permanently shut down its more expensive Grand Bahama tracking station. (The latter had provided launch support for over three decades, from 1954 to 1987, first for the Air Force and then for NASA.) Jonathan Dickinson already had everything that Goddard engineers were looking for, including radar, telemetry, a microwave relay to the Cape, and a command destruct system that could be remotely activated from the Cape if it were ever necessary to protect life and property should a launch go awry.

Since MILA was so crucial, the site continually evolved and was upgraded. The most dramatic change was its transition from a mostly human-operated site to autonomous operation, which has, not surprisingly, significantly reduced costs. While not a switchover to purely unattended (or "lights-out") operations, the change brought on by the ever increasing reliance on automation and computer processing has been beneficial, significantly reducing the station's staffing requirements. During the height of Apollo and for STS-1, for instance, the station employed upwards of 140 workers. That number has dropped dramatically to where less than 40 people are now required.[11]

Even as staffing was being reduced, modernization of technology increased. In 1995, the station went to an "all fiber" system, with fiber optics replacing all the communication lines between MILA, PDL and the control facilities at the KSC. A year later, a UHF voice system with a powerful, state-of-the-art quad-helix antenna was installed to support the Shuttle in the event of a Return to Launch Site (RTLS) abort. Today, Merritt Island has become a full-service spaceport communications facility, boasting a suite of 15 antennas that support all phases of a Shuttle flight—from prelaunch checkout to launch, on-orbit (via TDRSS) and landing. Leveraging each other's assets has enabled the DOD and NASA (and more recently the commercial launch industry) to support a wide range of space launches from Florida. As former Station Director Tony Ippolito put it, "All of this has allowed us a more business oriented approach in the operation of MILA."[12]

With near-Earth space communications now well covered by the TDRSS and the SN (with the Jet Propulsion Laboratory's DSN handling planetary work), the emphasis for NASA to support suborbital science missions has, in turn, made Goddard's Wallops Flight Facility home to the GN's most





Merritt Island, MILA (top) and Ponce de Leon, PDL (bottom) provided uninterrupted launch vehicle tracking out of the Kennedy Space Center (KSC). Shown are the MILA operations building along with the station's two 9-meter (30-foot) Unified S-band (USB) antennas used for tracking, telemetry, command (TT&C), and voice. The less complex PDL "wing station" had a 4.3-meter (14.1-foot) antenna used to cover loss-of-signal at MILA from the exhaust of the Shuttle's Solid Rocket Boosters (SRB). (Un-numbered Kennedy Space Center images, *science.ksc.nasa.gov/facilities/mila/milstor.html*, *science.ksc.nasa.gov/facilities/mila/pdl.html*, accessed 21 November 2005))

extensively equipped facility. Located on Wallops Island off the Delmarva Peninsula coast of Virginia, Wallops is NASA's lead facility for implementing its suborbital and special low-orbit research projects. Established by NACA in 1945, the 6,200-acre facility is today staffed by 1,000 full-time government personnel and contractors who support everything from sounding rocket and balloon launches to conducting unpiloted aerial vehicle research.[13]

The beginnings of Wallops date back to the end of the Second World War. In 1945, NACA authorized the LRC to develop the small offshore island into an aeronautical range where rocket propelled models can be launched to conduct studies of the upper atmosphere. In this way, Wallops became the oldest civilian launch site in the United States. The facility allowed Langley scientists to have many more options in conducting their research, like overcoming the limited capabilities of the wind tunnels of the day, for example. With the establishment of NASA in 1958, the creation of the "manned-satellite" (Mercury) program and Wallops's close association with Langley and its STG, much of the activities there quickly turned to developing the components needed for putting a human in space. This included designing capsule escape techniques, pressure testing of the early blunt-body aerodynamic designs and flight test support of heat shield development and ocean recovery techniques.[14]

In addition to the emphasis put on Mercury, research in the aviation arena continued. The facility's airport, for instance, was used to develop and test runway surface designs for aircraft noise reduction. And it was at Wallops that the Scout launch vehicle solidified its place in history as the premier rocket for launching small payloads for the scientific community, with a remarkable 100 percent success rate since 1976. It was here that the Scout became the first solid fuel rocket to place a satellite into orbit when, on 16 February 1961, it successfully launched a 44-kilogram (96-pound) NASA atmospheric research payload into orbit.[15]

On 19 October 1981, the Wallops Flight Center, as it was then called, was consolidated under GSFC management and redesignated the Suborbital Projects and Operations Directorate, otherwise known as the Wallops Flight Facility. Less than five years later, in April 1986, the tracking station that was part of the NTTF located on the grounds of GSFC, was transferred to Wallops. The flight facility now had the added responsibility for capturing small satellite telemetry, tracking, and command. Many of the first satellites supported from the facility would go on to become some of NASA's most successful orbital science platforms. Among them were the IUE, the Inter-planetary Monitoring Platform (IMP-7), and the Cosmic Background Explorer (COBE). To better handle the additional workload, the facility soon underwent a one-year modification where existing hardware was supplemented with equipment from former STDN stations around the world that were then being phased out. A new communications system was added as part

of the upgrade to transmit data from Wallops to the Project Control Centers located back at Goddard.[16]

In the late 1990's, the facility began developing ways to really expand its sphere of operations so as to more effectively support launches at locations away from Wallops Island and the immediate Virginia coast area. Mission operations at Wallops took on a new dimension when it began operating the Mobile Range Control System, or MRCS. Developed by the Center's Electrical Systems Branch, the MRCS is a self-contained, transportable launch system that can be loaded into a military cargo aircraft such as the C-130 and flown around the world to conduct satellite launches at remote locations as needed. It in fact acts somewhat like a transportable range, equipped with an Uninterruptible Power Supply, a range safety display and redundant command destruct transmitters for flight termination along with all the necessary computers and communication equipment needed to support a launch in a "turnkey" fashion.[17]

Before there was the MRCS, setting up a mobile range was much more cumbersome and logistically demanding, translating into higher cost. Equipment in several vans and trailers had to be transported either by air or by sea and put together upon arrival at the remote location. One former MRCS Project Manager noted the tremendous advantage this new system offered, saying "In comparison with the older collection of subsystems in separate trailers, the fully integrated MRCS can be completely tested prior to shipment. This helps reduce mission support and cost."[18]

True to its calling, the Wallops's mobile range has been well traveled since 1997, supporting launches from the nearby Coquina Outer Banks of North Carolina, to the Canary Islands in the East Atlantic and even as far north as Kodiak Island, Alaska. To support the commercial launch market, the MRCS was granted a license in 1999 by the FAA's Office of the Associate Administrator for Commercial Space Transportation (FAA/AST), which allows private paying customers from the U.S. commercial launch industry to use the system to launch their payload into space.[19]

All these developments have made Wallops Island (also known to the commercial launch sector as the Mid-Atlantic Regional Spaceport) America's preeminent small rocket facility. As a controlled range, it has the authority to clear airspace and reroute planes in times of need. Since Wallops's mission is so diverse, the ground station there is somewhat unique in that it has a combination of some very old antennas alongside state-of-the-art equipment. It still operates, for example, an original VHF antenna for ISS and Russian Soyuz voice support. A VHF Satellite Automatic Tracking Antenna/Satellite Command Antenna on Medium Pedestal—SATAN/SCAMP telemetry/command system—from the 1960s can also be found still operating there. Although rendered obsolete when stations began using microwaves to transmit data over long distances, this old system was kept to support the facility's suborbital and short-range data needs. Also, there are the original



The remote barrier-island location of Wallops Island on Virginia's Eastern Shore makes it ideal for testing aircraft models and launching small rockets. As the space program evolved, it became one of the Agency's mainstays for launching sounding rockets carrying scientific experiments into the upper atmosphere. In the 1980s, however, a proposal emerged to close Wallops as a way of reducing NASA's operating costs. Instead, officials decided to incorporate the facility into the Goddard Space Flight Center (GSFC) as it relied on the facility for satellite launch, tracking and data support. In this way, Wallops Island Station became the Wallops Flight Facility managed under the Suborbital Projects and Operations Directorate at GSFC. (NASA Image Number GPN-2000-001323)

9-meter (30-foot) S-band antennas that tracked Apollo astronauts to the Moon and back. These "antiques" can be found still being used everyday alongside the station's state-of-the-art 11-meter (36-foot) S-/X-band dual-feed antenna. As one NASA manager puts it, "They have practically one of every kind of antenna out there," which, in some ways, makes Wallops the perfect setting as a nostalgic rocket range.[20] It bridges the gap between an old fashioned test range nestled along the Atlantic coast and the modern twenty-first century spaceport.

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Satellites and spacecraft circling Earth today rely on both the SN and the GN in different ways. The GN of today is used primarily to support aeronautical and atmospheric research, range safety, and high inclination (high latitude) orbital communications. It is in this setting that the new era of NASA's communications network is found, the hallmark of which are *technology expansion* and *commercialization*.

First, the rapid expansion—indeed evolution—in digital telecommunications technology over the last quarter-century has made NASA's space communications a truly global amalgamation that connects every corner of the world. This same technological evolution has also greatly improved the ability of today's stations to perform TT&C functions compared to the previous STDN generation. Station autonomy has greatly reduced the requirements for human staffing. The objective is not to eliminate human-in-the-loop but let automation do what can be done in terms of scheduling, redundancy, and self-testing. Advancements in digital signaling and transmission techniques have allowed for the ever increasing demand for higher bandwidths (traffic) and lower bit-error-rates (accuracy) to be accommodated.

The other trait which can be used to describe the Agency's network operations is commercialization. This should not be surprising when one looks at the trend of space communication in which NASA has historically set the precedence but is now heavily influenced by the commercial sector. Just like the demand for better technology is always a driver, as space moves from the realm of government sponsorship to being a commercial commodity with increasing private industry participation, cost reduction,—and more importantly—profit in today's world of real-time global communications is more important than ever. It is these fundamental paradigm shifts that have taken NASA's STDN of the past to where it is today. This shift has enabled NASA in recent years to put ground stations in very remote regions of the globe where it was just simply not feasible a generation ago.

Take Antarctica, for example. The manpower that would have been needed to make a continuously operating ground station cost effective from such a location would have, in the past, been difficult at best. On top of that would have been the technical challenge of how one would get the data received at the station in a timely manner to their users who may be scattered across many continents.

In 1956, the U.S. Navy established McMurdo Station on the continent of Antarctica. At 77° 50' south latitude, McMurdo is well inside the Antarctic Circle and is the southern most harbor in the world. It is also Antarctica's largest community and the continent's center of activity. Built on the bare volcanic rock of Hut Point Peninsula on Ross Island, it is the farthest south solid ground that is accessible by ship. As early as 1901, McMurdo took on some sense of import when it became the staging point for the race to plant the first flag at the South Pole. Among the landmarks still preserved (by the New Zealand government) from that era is Hut Point, left behind by the doomed expedition of British Naval officer Robert F. Scott and his party in 1910. That year, Scott—with his team of four companions—embarked on an expedition with the aim of becoming the first man to reach the South Pole. The 2,964-kilometer (1,842-mile) trip was the longest continuous sled journey ever attempted in the polar regions. On 18 January 1912, they reached the

bottom of the world only to find the tent and flag of the Norwegian explorer Roald E. G. Amundsen, who had achieved the goal only five-weeks earlier. Demoralized and short on supplies, Scott and his men never made it back to McMurdo. The return journey ended in the loss of the entire party. Scott came to within 18 kilometers (11 miles) of a supply depot when he and his remaining two teammates perished of starvation and exposure. Their remains, along with diaries left by Scott in his tent, were found by a search party almost eight months later.[21]

Since 1956, McMurdo has grown from an outpost of a few buildings to the largest community on the icy continent with more than 100 structures, an outlying airport (Williams Field) with landing strips on sea ice and shelf ice, and a helicopter pad. Despite its remote location, McMurdo is among the most ethnically diverse communities per capita anywhere to be found. During the summer months, the population can swell to over 1,000 people, attracting scientists, construction workers and polar explorers from all nations around the world. During the harsh winter months of March to October, the population usually drops down to below 250 people who, except for time of emergencies, find themselves pretty much isolated for the winter.

Like a small town, there is a freshwater system, sewer, telephone, and power lines linking the buildings. Science equipment at McMurdo include diving equipment, recompression chambers for diving accident victims, cosmic ray monitors, and facilities to study magnetosphere and ionosphere phenomena. From the runways of Williams Field 16 kilometers (10 miles) away, flights span the continent and to airbases in and out of New Zealand. While skid-equipped planes can fly in and out of the frozen landing strips year-round, it was not until 1992 when a permanent, hard-ice runway on the Ross Ice Shelf was completed that larger transporters equipped with wheeled landing gears could come and go more frequently thereby greatly increasing the availability of supplies to the delight of the personnel stationed there.[22]

It is in this unique part of the world that NASA teamed up with the National Science Foundation (NSF) to establish the southern most satellite data acquisition station in the world.[23] The McMurdo Ground Station (MGS) is today home to a 10-meter (32-foot) S- and X-dual band NASA antenna located atop the 152-meter (500-foot) Arrival Heights peak. From this vantage point, it has a fantastic view in all directions. Looking south, it can see satellites on the other side of the Pole. NASA's original requirement there was to support a joint effort by the two Agencies (along with international partners) to radar map the entire Antarctic continent by satellite.

Operational since 1996, MGS started out collecting X-band telemetry (frequencies in the 5- to 11-gigahertz range in the electromagnetic spectrum, higher than C-band but lower than Ku-band) on about 25 passes each day from ERS-1 and ERS-2, the European Earth Resource Satellites, and the Canadian synthetic aperture radar mapping satellite *RADARSAT*. The



McMurdo, Antarctica is the world's southern-most port and home to numerous expeditions to the South Pole since 1901. The McMurdo Ground Station is located on nearby Arrival Heights Peak. (Photograph courtesy of NASA)

station's S-band capability was put to use not long after in August of 1997 when a NASA Lewis land imaging satellite malfunctioned and began tumbling shortly after launch. Because MGS could see virtually every pass, it was a real asset in the rescue attempt. Unfortunately the spacecraft could not be stabilized and was consumed in a fiery reentry just 36 days after its launch. NASA today uses McMurdo as a data collection hub for satellites monitoring ice movements in the Southern Hemisphere. Such data is used immediately on site by scientists on the continent as well as by those planning re-supply shipping routes in and out of Antarctica.[24]

Despite the fact that few other ground stations have the capability of MGS to collect the enormous volume of data that can only be done at the Poles, communications in and out of the continent is still not so good. The only way to get data out is through something called the McMurdo TDRSS Relay System, or MTRS, which consists of two antennas (4- and 7-meter [13- and 23-foot] dishes) that actually communicate with the TDRSS. NASA uses a nearby microwave tower to relay signals to the MTRS at a place called Black Island located about 50 kilometers (30 miles) closer to the Equator. Due to the curvature of Earth and the way a satellite travels in polar orbit, near the poles, even this relatively short distance can make a big difference to provide

a much better view to relay satellites. The drawback for NASA, however, is that since the link is shared with the NSF, it is only used occasionally so as not to overwhelm the NSF's ability to send data off the continent on the always busy TDRSS.

Rounding out Antarctic communications is a small system located right at the South Pole. Although not really part of NASA's GN, it allows polar scientists there to communicate with TDRS-1—the original satellite—on brief occasions when it pops above the horizon while performing its "figure-8" loop in the vicinity of the Equator. These ongoing efforts to build a good communications network on this most desolate of places has only recently culminated, allowing the inhabitants to join that most global of communities: the Internet. This accomplishment is not lost on those who run the space agency's networks. The proclamation "We brought internet to the South Pole!" sums up the Agency's legacy on Antarctica rather nicely.[25]

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Antarctica and the South Pole are not the only places to have been "tamed." In this new era of ground stations, NASA has also been busy on the



A team from the Goddard Space Flight Center (GSFC) visit the Intelsat communications relay station on Black Island in 1999. Note the microwave tower link back to McMurdo. (Photograph courtesy of NASA)

other end of the globe. While the mid-latitude location of the continental United States makes for a good setting for launching science payloads into orbit (Wallops, Cape Canaveral), it cannot however, provide routine, low-cost, launch access to investigate interesting activities that permeate the upper atmosphere in the polar regions, activities such as Aurora Borealis, or the northern lights. To do this, one must venture near the Arctic Circle, to a place called Poker Flat some 30 miles north of Fairbanks, Alaska.

Owned and operated by the University of Alaska's Geophysical Institute since 1968, Poker Flat Research Range has been primarily dedicated to the launch of sounding rockets for the purpose of middle to upper atmospheric research. The first rocket was launched there in 1954. The rather enticing name Poker Flat is believed to have been taken from American author and poet F. Bret Harte's rags-to-riches short story, *The Outcasts of Poker Flat*, which in a way describes the inauspicious beginnings of the original ad hoc launch site that was constructed from begged and borrowed materials. But the range could have simply been named after nearby Poker Creek. In any case, Poker Flat is today the only nongovernment, university owned and operated range in the world. It is also the only high-latitude, polar region, rocket launching facility in the United States.

Because of its importance, NASA has funded the operation of the range under a cooperative agreement with the University of Alaska's Geophysical Institute since 1979, assuming funding responsibilities previously held by the National Oceanic and Atmospheric Administration (NOAA). Much bigger than the birthplace of America's missile activities at White Sands, Poker Flat is in fact the world's largest, land-based rocket range. It consists of a chain of downrange flight and observation sites spanning inland Alaska to Spitsbergen in the Arctic Ocean that are used to monitor and help recover payloads. Since it is an active rocket range, NASA and the university have to coordinate their activities with many U.S. government agencies. The FAA must approve and coordinate the air space during launches. Also, since the range is so large, permission to impact rockets and their payload on its 26 million acres of land has to be authorized by a whole host of government agencies, including: the Bureau of Land Management; the U.S. Fish and Wildlife Service; the State of Alaska Division of Lands; Doyon, Ltd. (the largest private landowner in the state of Alaska); and the Village Traditional Councils of Venetie and Arctic Village. Unlike bygone days at the dawn of the Space Age, environmental regulations mandate much of what can go on at these ranges today.[26]

As with the oil pipelines a quarter-century before, Alaska today serves as the great northern frontier. But this time, instead of energy, the commodity is information. More specifically, the information age revolution and commercialization of space. In fact, AGS, the Alaska Ground Station at Poker Flat, is not really even a NASA owned station at all. Rather, it is part of a commercial network of ground stations called DataLynx, which is owned and

operated by Honeywell Technology Solutions, Inc. With U.S. and international partners such as Universal Space Network, the Australian CSIRO, and the Japanese Institute for Aerospace Technology (JAXA), just to name a few, DataLynx has today expanded to over 20 ground stations on six continents.

Mirroring in many ways NASA's STDN of the previous generation, DataLynx stations today operate in a latitude band spanning 78° north at Svalbard, Norway, down to 33° south at the Santiago Station operated by the University of Chile. In fact, locations of many of the stations harbor back to the STDN days (and even earlier), with places like Hartebeesthoek, South Africa; Perth, Australia; and as mentioned, Santiago. Other places such as Beijing, China—a location which would have been impossible to imagine during the Cold War—have become part of this new age in commercial networking designed to serve satellite-using customers from around the world. With profit openly the bottom line, DataLynx, which depends highly on automation and "lights-out" operations, advertises itself as a " rapid, proven, reliable, cost effective mission-critical . . . distributed partner network," one that offers 24/7 command, control, and communications for "broad and flexible solutions, reducing cost and risk to our clients so that they can focus their resources on their core businesses."[27]

One of these clients is none other than NASA, the one who subsidizes Poker Flat, the very range that the Alaska Ground Station sits on. With $5 to $6 million a year, the Agency literally buys a minimum number of passes each day using the station's 11-meter (36-foot) antenna. (AGS also supplements this system with a smaller 5-meter (16.4-foot) S-band system called the Low Earth Orbit Terminal as well as a somewhat larger 8-meter (26.24-foot) transportable S-band antenna called the Transportable Orbital Tracking System.) Assuming 36 passes a day, this averages out to approximately $400 per pass, a figure that is much more economical to NASA than what it would cost to otherwise engineer, build, operate, and maintain a station of its own. It is therefore a truly joint government/industry arrangement that in the end benefits both NASA and the DatyLynx stakeholders.

Program Executive Bill Watson explained the arrangement which in a way captures the business-end of how ground stations in the twenty-first century works: "In this case, Honeywell not only owns DataLynx but they also won the current contract for operating the GNs. So they consolidated the deal and said we will treat the government assets and commercial assets as a pooled resource. You give us a guaranteed annual amount of revenue and we will guarantee you a minimum number of passes per day, and then if we [NASA] go over that, we pay by the pass. So it's a quasi-government commercial activity."[28]

While the Alaska station at Poker Flat is located just outside of the Arctic Circle, there is yet another ground station which continues this joint, government-to-private enterprise theme but is situated a mere 965 kilometers (600 miles) from the North Pole itself. It is here on the Norwegian archipel-

ago of Svalbard that the world's northern-most, permanent, satellite ground station can be found at 81° latitude near the top of the world.

The earliest written records documenting the existence of these frozen polar islands date back to the late twelfth century by the Vikings as they sojourned about the Arctic Ocean. For the next 400 years, though, Svalbard—which means "cold edge" or "cold coast"—was largely forgotten. Then in 1596, the islands were accidentally rediscovered by an expedition led by the great Dutch seaman Willem Barents while searching for a Northeast trade passage to Asia. This was followed by the English explorer Henry Hudson, who mapped the area and reported good whaling there. This spurred a bitter quarrel between English and Dutch whalers over the territory. In 1618, a compromise was reached, with the Dutch limiting their operations to the northern part, leaving the rest to the English, the French and the Hanseatic League (an alliance of trading cities that maintained a trade monopoly over northern Europe between the 13th and 17th centuries). The Danes also claimed the archipelagos as part of Greenland.

Over the next 300 years, various countries such as Norway, Russia, and Sweden laid claim to the islands, this especially after coal—the great source of energy that could empower the new steam engines—was discovered there in the late nineteenth century. Norway finally took formal possession of Svalbard in 1925 after a treaty was signed in Paris after the First World War. (Russia, who did not sign the accords, was to dispute Norway's stance on the islands well into the latter part of the century.) The islands again came to the forefront when, during the Second World War, it was the scene of some very intense naval battles between Germany and the Allies due to its rich deposit of coal.

Although Svalbard's claim to fame for years had been its geographical setting serving as the staging point for North Pole explorers, it was not until 1990 that Norway officially opened up the region to general tourism, greatly expanding its economic base and spurring on the development of new industries. In 1997, the Norwegian Space Center (NSC), along with the private space conglomerate Konnesberg Satellite Services (KSAT), began putting together the Svalbard Satellite Ground Station at Platåberget, near the town of Longyearbyen. SvalSat, or SGS, is today a truly general purpose facility, providing customers with tracking, telemetry, and data returns from a host of polar orbiting satellites.[29]

While NASA knew that the Alaska Ground Station was in a fairly good location at 65° latitude, it could, nevertheless, only observe about 14 out of every 16 passes that were actually available each 24 hours. With SGS being less than 1,000 kilometers from the North Pole, it could literally see every single polar orbit pass. Therefore, when Norway approached NASA to join their operations, it was a rather easy decision. The Agency first put up a trailer (often covered by a tent to keep the snow off) and an antenna in 1997. The site has steadily grown since then into sort of a "space park" of the Arctic. NASA now



With its extreme northern location on the Svalbard archipelago (78°13' latitude), SGS is the only ground station in the world able to provide all-orbit support of polar orbiting satellites. Six multi-mission antenna systems, along with several minor antenna systems, are used for TT&C and operations. One of the systems is dedicated to the NASA Ground Network (in shared operations with EUMETSAT) and is operated locally by SvalSat. The remaining antennas are remotely controlled and operated from Tromsø at the Tromsø Network Operations Center 1,000 kilometers (600 miles) to the south on the Norwegian mainland. (KSAT photograph, *https://www.spacecommunications.nasa.gov/spacecomm*, accessed 22 August 2007)

operates an 11-meter (36-foot) S/X-band antenna there. Here, the Americans join the Norwegian Space Center and Kongsberg, as well as EUMETSAT (the European Organization for the Exploitation of Meteorological Satellites), to make up the world's northern-most tracking station.

Surrounded by "the King of the Arctic" polar bears—personnel are required to carry a weapon when working outside—the space park continues to grow. One of its hallmarks is the ability to get large amounts of data quickly off the island with the use of fiber optics. NASA arranged in 2004 to have the NSC install redundant fiber optics all the way back to the United States, much to the delight of data users on the North American continent. With this capability, NASA's communications to Norway today are actually much more robust than those to Alaska. Taking the Agency's domestic, government-industry relationship in Alaska to an international level, the operation in Svalbard serves as a model for the way space is being treated openly as an international commercial commodity.

Unlike days bygone, the United States and the (former) Soviet Union are by no means the only torch bearers. Watson explained:

> The Norwegians made us a deal. They said you're [NASA] paying $6 million *a year* now for commercial relay satellite services to haul data out of Norway. For $5 million for *five years*, we will install the fiber and then give you the next *15 years* for free. How could we turn that deal down! And what they did was they made a similar offer to NOAA and so between NASA and NOAA, they got a revenue stream of about $10 million a year. They went off to a commercial financier and got the money, . . . basically borrowed it against a promissory. The implication was that NASA and NOAA would commit to pay for five-years. . . . They brought two ships in and laid redundant fiber, two different paths so if one gets cut, we still have the other. It was a remarkable plan.[30]

Svalbard has proven to be a win-win situation for both NASA and the Norwegians. What began as a joint venture to gather a few more satellite passes has basically turned this faraway mining town into what is one of the best wired and well connected places anywhere in the world. In a way, NASA's operations at Svalbard bring the GN of the twenty-first century full circle to how it all began nearly half a century ago. Back then, the emphasis was on hiring local people and training them to "nationalize the station." This worked well at many places, from Chile to Ecuador to Guam. With Norway, this is very much a continuation of that legacy, except NASA no longer has to provide all the technology and set all the precedence. At Svalbard, the Norwegians did not need that "leg up" to turn their concept into a reality.

In a sense, while America was winning the space race, the rest of the world caught up.

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Throughout the 1990s, the phrase "Faster, Better, Cheaper" became somewhat of a choreographed aphorism that drove much of the way business was conducted in the high technology world, both in the private sector and in the government. This approach impacted the space program in ways ranging from economics to performance and, some would argue, safety. For critics, "Faster, Better, Cheaper" was usually followed by "Two out of three aren't bad!"

In this era of commercialization, the approach as to how ground stations were to be built and how they were to be used began to change. "Lights out" operations and station autonomy entered the scene. Companies like USN and DataLynx entered the playing field. These multimission network terminals offered users the advantage of low cost services based on the philosophy of "pay only for what you use." Like Santiago, they provided services on a retainer basis with added "per-pass" cost on actual usage, targeting not only commercial users of satellite services but also government users like NASA.

It was in this renewed atmosphere of cost awareness and infrastructure reduction that NASA tried implementing an across-the-board streamlining of its organizations. This included, in particular, its tracking and space communications operations. The goal was actually rather sweeping but straight to the point: back the government out of day-to-day operations. To do this, all management of space operations was consolidated under a single office at the JSC in Houston. The office was named the Space Operations Management Office. Known as SOMO, the name would seemingly take on a life of its own in the coming years, the mere mention of which, even today, conjures up strong feelings on the part of those who were involved.

The decision to establish SOMO was prompted by an Agency-wide examination of space operations requirements as part of a so-called Zero Base Review, completed in 1995. In this review, representatives from NASA Field Centers evaluated opportunities for consolidation, privatization, and commercialization of existing government functions across NASA. The review team recommended several initiatives to achieve cost savings while ensuring a continued, high quality of operations services. It was decided that a single but consolidated, management structure could be implemented that would best accommodate these goals. SOMO would be that management organization. Theoretically, it could be a centralized office that could quickly respond to service requirements as identified by specific NASA programs and projects, and even to external (non-Agency) customer requests for similar services—as long as NASA operations were not interfered with. Whether this was a good idea or not, at least on paper it seemed like it could work.[31]

Under Administrator Daniel S. Goldin's direction, the SOMO was established the following year. Its central objective was to ensure that existing NASA assets were used as efficiently as possible and that duplication was avoided. This objective unavoidably resulted in shuffling of responsibilities (and power) between various organizations within NASA. Not only that, it would also go one step farther by *eliminating* certain offices.[32]

SOMO was purposely designed to be small, with key positions held by about three-dozen individuals from NASA Field Centers who, for all intents and purposes, made the decisions in carrying out all the Agency's space operations. Specifically, the Data Services Manager was from the JPL; the Missions Services Manager was from Goddard; and the Commercialization and NISN Manager was represented by the MSFC.[33]

Thus from JSC, the office soon ended up basically managing all of NASA's space operations, including its vital communication networks and tracking systems. The TDRSS Space Network became a part of it as well as the GN, the NISN and JPL's DSN. In other words, management of the networks suddenly—and for those involved, unbelievably—now came under the auspices of the JSC. This gave JSC an inordinate amount of power. While the move did not take away the GSFC's day-to-day responsibility of operat-

ing the network (nor JPL for its Deep Space activities), it did significantly erode the role that Headquarters had in Washington. The Office of Space Communications (OSC, organizational designation "Code O") responded unreservedly to the sweeping mandate to, in effect, do more with less. Staffing at OSC was first reduced by 35 percent, then by 60 percent, and finally, 85 percent. This included elimination of 16 senior level GS-15 and higher positions. With its role greatly reduced, the job of Headquarters was relegated to conducting external interface, determining program requirements and strategic planning.[34] In short, Code O was one of those, which for all intents and purposes, eliminated. The numbers back this up. Prior to consolidation, Code O had in excess of 50 people at Headquarters working the Agency's tracking and communications needs. After the reorganization, scarcely eight remained.[35]

To put it bluntly, space communications—at Headquarters as well as Goddard—was being gutted.

The idea of consolidation was radical but seemed noble enough at the time. NASA established the SOMO (Code M) to oversee its space operations activities and to implement a single Consolidated Space Operations Contract (CSOC) as the initial step to reduce the cost of operations. In this "integrated operations architecture", many of the trends which were permeating space communications in general—trends such as the aforementioned automation and privatization—could take effect. The thinking was that a single, large contract would naturally be more efficient than a plethora of smaller ones. On 25 September 1998, NASA awarded the enormous $3.4 billion operations contract—five year base with an option for an additional five years—to a team led by Lockheed Martin. It was one of the most valuable outsourcing programs ever undertaken by a civilian agency. Under CSOC, five contracts which had up until then operated independently were consolidated on the first day that the contract took effect. (Ten more separate contracts transitioned to CSOC from 1999 to 2004.)[36]

Even NASA was fully aware, however, that the envisioned cost reduction could not take place overnight. While some reduction could be gained initially, the contractor work force supporting space operations at five NASA Centers would have to be reduced gradually over the 10-year period at a rate of slightly less than 100 jobs per year on the average. The idea was that, by implementing the reductions over a decade, essentially all of the attrition could be absorbed through planned retirement, personal job changes and reassignment of contractor employees to other programs.

Even as CSOC was being awarded to industry in 1998, SOMO began exploring several additional commercialization initiatives with the aim of realizing some further, longer term cost savings. For example, services with USN were established. Another component was to provide, using the CSOC contract-vehicle with private industry, opportunities to offset some of NASA's operating cost by marketing unused capacity on the TDRSS. One such ini-

tiative—originally conceived by Code O—was to provide "TDRSS time" to commercial oil exploration vessels at sea, as there was virtually no commercially available communication satellite that could support the transfer of the large amounts of data for such application. Through these and other initiatives, SOMO at the JSC projected that a $1.4 billion cost saving could potentially be realized over the next 10 years.[37]

As optimistic as JSC was on the outside, however, it could simply not shed a barrier that soon (and clearly) manifested itself as a growing cancer in the whole SOMO idea. When the office was formed in 1996, program responsibilities moved from Washington to Houston. This had huge repercussions. While the various Field Centers still had their programs to work, this shift in responsibility to Houston naturally did not sit well with them. According to Robert Spearing, who witnessed the whole thing and was one of those actually recruited back from private industry by NASA to help fix the problem, it was putting in charge "an organization that they [the Field Centers] felt was ill-equipped to deal with their issues." Said Spearing,

> What was not understood well was what happens to the work that has to go on at Headquarters is very difficult for a Field Center to perform because they don't have the skills for that. A lot of our work here at Headquarters relates to working with the other agencies of the federal government, both civilian and military, and also working with the Congress to advocate our programs. So you ended up having a very limited capability to perform that function here at Headquarters.[38]

Essentially, SOMO had grossly underestimated the importance of having a team in Washington to take care of business with the rest of the federal government (and international partners). Said Spearing:

> You have to have the right talent here in town for our relationships internationally. . . . When you work with international organizations like the European Space Agency or the Italian Space Agency or the Japanese Space Agencies, all of these organizations look to NASA Headquarters. When someone from a Field Center goes and represents NASA to these organizations, they see that as somewhat of a mismatch. So there were some lessons learned.[39]

By 1998, just two years after SOMO was set up, it was already clear to most that in the process of trying to work issues between the JSC, Headquarters, and the other Field Centers like Goddard, Marshall, and JPL, that it just was not working. Space Communications had been relegated to an office under Code M. There were open and often ugly struggles over who was



NASA's management of its tracking and space communications activities began at Headquarters with the establishment of the Office of Tracking and Data Acquisition (OTDA) on 1 November 1961. This photograph shows its staff in the early 1980s. OTDA was reorganized into the Office of Space Tracking and Data Acquisition (OSTDA) in 1983 and then into the Office of Space Operations (OSO) in 1987. It became the Office of Space Communications (OSC) in 1992. Edmond C. Buckley was OTDA's first Associate Administrator (1961–1968). He was succeeded in that position by Gerald M. Truszynski (1968–1978), William C. Schneider (1978–1980), Robert E. Smylie (1980–1983), Robert O. Aller (1983–1989), Charles T. Force (1989–1996), Wilson T. Lundy (interim Deputy AA 1996), David W. Harris (interim Deputy AA 1997–1998), Robert E. Spearing (1998–present). (Photograph courtesy of Charles Force)

in charge and who had control. Beset by quagmire, something had to be done. The SOMO in Houston eventually came around to this realization. A rather laborious process followed in which the OSC was slowly restored. First, the position of Deputy Associate Administrator for Space Communications under Code M was abolished. It was then placed under the Office of the Deputy Associate Administrator for the Space Shuttle. Later in the year, it was again transferred, this time under the Director for Resources Management. By 2001, Stan C. Newberry, who by then was heading up the SOMO, was holding talks about the future of the office with top agency officials at Headquarters. Not soon thereafter, and with little fanfare, Administrator Dan Goldin dissolved the controversial office and restored program management of space operations back to Washington where they remain today. With the restoration of Code O, management for tracking, data acquisition, and space communications at NASA

were reconsolidated. Ten years after the controversial reorganization began, the OSC was finally restored to the structure Buckley had setup 45 years earlier.

In April 2004, at the end of the five-year base period, the CSOC industry contract to the Lockheed-Martin team was terminated and not exercised into the option period.[40] OSC at Headquarters not only survived the SOMO fiasco, it broke apart CSOC. The separate set of five contracts (now collectively called Space Mission Communications and Data Services) divided CSOC's functions fairly equally across three NASA centers—Goddard, Kennedy, and Marshall. The new contracts were collectively worth about $400 million a year, roughly equal to the annual average of the old CSOC. But now Headquarters intended to award, manage and determine the specifics of each contract separately.[41]

Regarding the whole affair, James Costrell of the OSC said, "It was a new concept to NASA. The theory was that spacecom (space communications) is spacecom, that it's all the same. So NASA charged ahead with the idea that a consolidated contract would result in efficiencies."[42]

Of course, it didn't quite work out that way. With the contract managed by SOMO from Houston, other Field Centers found that Lockheed-Martin and its subcontractors could not respond to their needs very effectively. The contract had other weaknesses too. Among the worst was that NASA had to renegotiate prices with the contractors whenever conditions changed. For instance, if the Agency ended a mission, closed a tracking station or took any action that altered the work needed from the industry team, Agency and company managers had to agree on cost revisions. Through the lessons of SOMO and consolidation, NASA learned (the hard way) that one size does not always fit; divide and conquer sometimes works better. "What the agency finally came to grips with was that there are some fundamental differences between the various communications activities," Costrell said.[43]

Although the controversy and subsequent fallout of what is now simply referred to as "consolidation" inside NASA may have in many ways been a failed experiment and a bitter pill to swallow (many were reassigned from GSFC and Headquarters or left the Agency), the ensuing years have shown the resiliency of America's space agency as an organization to overcome and move forward from its setbacks.[44] The revolutionary, new kind of network ushered in by TDRSS and the new mission of today's network of ground stations have helped set the stage for the coming decade. On 14 January 2004, President George W. Bush announced his "New Vision for Space Exploration" to return astronauts to the Moon by the year 2020, to be followed by mankind's first journeys to a neighboring planet.

Space communications have indeed come a long way since engineers first tried to keep track on a little sphere called Sputnik as it beeped its way around the globe. Today, that same technology—which allowed the world

to watch live broadcasts of the 1964 Tokyo Games to the "space handshake" of Apollo-Soyuz and webcasting of telemedicine from Antarctica—continues to provide people around the world with sharing of everyday technologies not possible before. Such is the diversity that has allowed live communications from virtually anywhere in the world. Families who have not seen each other in years can stay in touch using cellular telephone networks and video networking on their personal computers. Brilliant HDTV via satellite can be enjoyed, for example, by the outdoorsman on a camping excursion hundreds of kilometers from the nearest city.

As NASA prepares to send humans back to the Moon and launch evermore ambitious space missions into Earth orbit and beyond, the tracking and communications network which Jack Mengel, Ozzie Covington, Ed Buckley and so many others began not so long ago will be there to meet the challenge.